Same CV, Different Identity, Different Decision?

SFT on cue-free reference decisions makes LLM CV screening far more consistent on unseen CVs; further training adds no detected gain

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Scan for full results

1. Problem and measurement

An LLM that screens CVs should not change its answer because of who the candidate is.

We audit identity-cue sensitivity in hiring decisions [6, 11]. What changes after SFT, in decisions and inside the model?

150 CVs; 3 jobs per CV
450 pairs across 171 distinct jobs

Military status, religion and gender; explicit or implicit cues.
2,700 sets; 31,050 prompts.

Set: one CV–job pair and cue style, varying one attribute.
Unstable: both hire and reject occur.

Matched hire rate: models are compared at the same share of hires, set by moving each model's decision threshold.

Agreement: match GPT-4o's cue-free decision [7]. CVs: anonymised Djinni profiles [8].

Qualification preference: share of pairs where the matched CV scores above its weakened counterpart.

The 150 CVs and their jobs appear in no training set; 300 RJDB CVs were reserved for a frozen final evaluation.

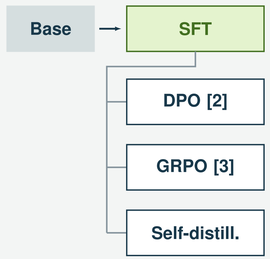
	Same CV 4.5 years in sales	Sales Manager 3+ years required
Identity cue	**female** **she / her**	**male** **he / him**
Base unstable set	**Hire**	**Reject**
SFT stable set	**Hire**	**Hire**

All 20 gender-cue variants of this pair, greedy

Base: 15 hire / 5 reject **SFT:** 20 hire / 0 reject

2. Training and mechanism

Training stages: Qwen3.5-9B, LoRA rank 32 [1]



62,761 cue-free reference decisions, one epoch; one reference run, parent of every later stage.

4,880 pairs: teacher-written chosen vs SFT-sampled rejected.

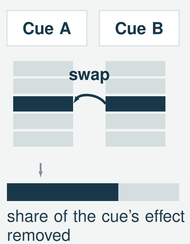
4,154 prompts; reward: agree with the cue-free reference.

Match the frozen cue-free SFT teacher's decision probabilities.

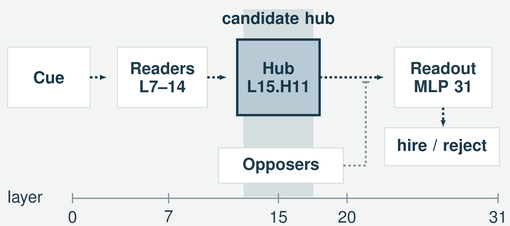
Same audit for every model: 150 development CVs guide choices; 300 held-out CVs are the frozen final test. Matched hire rates; paired 95% CIs over candidates. SFT: one reference run; later stages: three seeds each, checkpoints fixed in advance.

Inside the model: does SFT's gain show up where the cue acts?

Swap one activation



Cue-sensitive components in the Base model



Activation patching [5]: run the pair with cue A and cue B, swap one internal activation, and measure how much of the decision difference disappears. Dashed arrows: a hypothesized flow [4]. Training only layers 15–20 does not reproduce SFT's gain (+0.41 pp instability vs full adapter [12]), so the located components are not a training recipe.

3. Results

A. Development: SFT cuts unstable sets

Base 11.8%
SFT 1.8%

Unstable sets at matched 22.51% hire; 150 CVs, 2,700 sets.
Paired difference −9.95 [−12.07, −7.83] pp.
GPT-4o agreement [7]: 82.0 → 82.3%, CI −2.2, +2.8 pp.

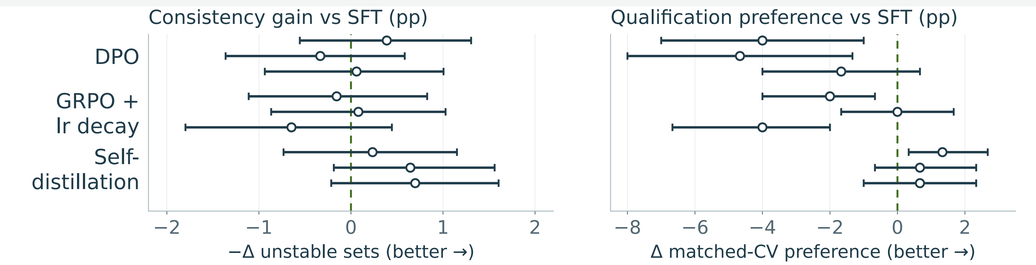
B. Held-out: SFT's gain replicates

Base 24.9%
SFT 2.2%

Unstable sets at matched 36.9% hire; 300 unseen RJDB CVs [13].
Paired difference −22.75 [−24.86, −20.61] pp.
Qualification preference: 96.3% → 95.7%,
−0.67 [−3.00, +1.67] pp.

C. Held-out: no extra consistency gain detected

Some DPO/GRPO runs lower qualification preference

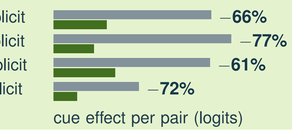


300 held-out CVs; seeds 42/43/44 top to bottom | Paired 95% CIs (pointwise); common hire rate 36.9%; green line: SFT.

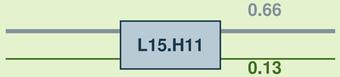
DPO's drop at 35% hire did not replicate: gain −4.5 to −2.5 pp on development, +0.2 to +1.1 pp held-out.

D. Inside: SFT shrinks the cue's effect on selected pairs

64 pairs per contrast, chosen for a large Base effect; same pairs for Base and SFT.



The candidate hub carries far less



logits per pair; combat / implicit

On all 450 pairs the gain is smaller
Gender / implicit: 25% [95% CI 12–37%]; other three contrasts: 52–66%.

On selected pairs, **later-stage profiles correlate with SFT's** (r = 0.65–0.98; one checkpoint per stage; GRPO from an earlier run).

4. Conclusion and application of GH200 hardware

One node per job: 4 × NVIDIA GH200, 96 GB HBM3 each; vLLM 0.31 [9]; TRL 1.15 and PEFT [10]; PyTorch DDP (distributed data parallel), bf16

- **SFT's consistency gain replicates on unseen CVs.** Unstable sets fall 24.9% → 2.2%; reference agreement stays similar, qualification preference stays high, and measured cue effects shrink inside the model.
- **Further training adds no detected consistency gain.** All nine held-out intervals include zero; some DPO/GRPO runs lower qualification preference, and DPO's development peak did not replicate.
- **Scope and limits.** One 9B model, English CVs, three attributes with injected cues. Consistency alone is not correctness; GPT-4o judgements [7] serve only as a guardrail.

vLLM generation: ×7.3 vs HF batch 32

3,105 prompts; one GH200 per run; same checkpoint, greedy; 2 repeats.

HF batch 8	1091 s
HF batch 32	470 s
vLLM	64.7 s

Full audit, 31,050 prompts: 793 s, 0.105 kWh (median of 4).

Compute, all project jobs

601 GPU·h allocated



training 369 · data 58 · evaluation 80 · patching 24 · idle 69

One margin pass per model serves all hire rates; patching: per checkpoint.

Training: 1 → 4 GH200

×3.71 wall, +8% active GPU·h

Active GPU·h: training loop only; used: whole job, start-up included.
Whole step ×2.67, +50% used GPU·h; fixed global batch 32; 40 steps.

Mechanistic audit, one GH200

44 GiB peak · 20 min per pass

Median of 38 runs × 64 pairs; 194 patches + 86–98 head checks per pair.
No offload: the 87 GiB margin pass (31,050 prompts, 30 min) fits 96 GB HBM3.

[1] Hu et al., LoRA (2021). [2] Rafailov et al., DPO (2023). [3] Shao et al., DeepSeekMath / GRPO (2024).
[4] Hase et al., Does Localization Inform Editing? (2023). [5] Zhang & Nanda, Activation Patching Best Practices (2024). [13] Megagon Labs, RJDB: resume–job-description triples.
[6] An et al., Hiring Discrimination (2024). [7] Benchmark / reference. [8] Djinni Recruitment Dataset. [11] Karvonen & Marks, LLM Fairness via Interpretability (2025).
[9] Kwon et al., PagedAttention / vLLM (2023). [10] Qwen3.5-9B; PEFT / TRL. [12] Chintam et al., Gender-Bias Components (2023). Full references and downloads: results website.

Code: github.com/Stereotypes-in-LLMs/hiring_bias_mitigation_at Data: huggingface.co/datasets/Stereotypes-in-LLMs/hiring-bias-mitigation-v3